

Computing the interleaving distance between two interval modules

Christian Chavez

I found nowhere the explicit computation of the interleaving distance between interval modules, though it is well known that it equals the bottleneck distance.

In this block I want to show

1 Preliminaries

We take $\mathbb{R}$ as the category induced by the poset $(\mathbb{R}, \leq)$.

1.1 The category of persistence modules

A persistence module is

An interval module is a functor $\mathbb{R} \rightarrow \text{Vect}(\mathbf{k})$. Let $\mathbf{k}$ be an arbitrary field, fixed throughout.

1.2 The shift functor

1.3 The interleaving distance between persistence modules

A t -interleaving of U and V is a pair of morphisms (ϕ, ψ) that make the diagrams

$$\begin{array}{ccc} U & \xrightarrow{\mathbb{1}_U^{2t}} & U[2t] \\ & \searrow \phi & \nearrow \psi[t] \\ & V[t] & \end{array} \quad \begin{array}{ccc} V & \xrightarrow{\mathbb{1}_V^{2t}} & V[2t] \\ & \searrow \psi & \nearrow \phi[t] \\ & U[t] & \end{array}$$

commute. If such a pair exists, we say that U and V are t -interleaved.

Définition 1. The interleaving distance between two intervalmodules is

$$d_T(U, V) = \inf \{t \geq 0 \mid \text{there exists a } t\text{-interleaving of } U \text{ and } V\}.$$

2 The computation